

```

16 //I have used a linked list implementation for history
17 struct list_elem {
18     struct list_elem *prev;
19     struct list_elem *next;
20 };
21
22 struct list {
23     struct list_elem head;
24     struct list_elem tail;
25 };
26
27 struct history_line {
28     struct list_elem elem;
29     int line_num;
30     char line_str[MAX_LINE_LEN];
31 };
32
33 //History list (It is global)
34 struct list history;
35 int history_index = 1;
36
37 //String functions made for xv6
38 int my_strncmp(const char *s1, const char *s2, int n) {
39     while(n > 0 && *s1 && (*s1 == *s2)) {
40         s1++;
41         s2++;
42         n--;
43     }
44     if(n == 0)
45         return 0;
46     return (unsigned char)*s1 - (unsigned char)*s2;

```

I implemented a linked list implementation for xv6, so I can store history upto last 10 commands, incorporating functions, calls and all within the code.

```

167 void history_show(void) {
168     struct list_elem *e;
169     struct history_line *hlp;
170     int total = 0;
171     int count = 0;
172
173     //Counting total commands
174     for(e = history.head.next; e != &history.tail; e = e->next)
175         total++;
176
177     int start = total > 10 ? total - 10 : 0;
178
179     for(e = history.head.next; e != &history.tail; e = e->next){
180         hlp = list_entry(e, struct history_line, elem);
181         if(count >= start)
182             printf("%d %s\n", hlp->line_num, hlp->line_str);
183         count++;
184     }
185 }

```

It shows the stored commands when they are parsed through getcmd into a buffer. The history_show() function is created to help the commands traverse in a list and they are displayed, when we write \$history in QEMU.

```

void history_run(char *buf) {
    struct list_elem *e;
    struct history_line *hlp;
    int index = 0;
    int found = 0;

    // Handle !!
    if(buf[0]=='!' && buf[1]=='!' && buf[2]==0){
        // Get last command
        if(history.tail.prev == &history.head){
            printf("-sh: no previous command\n");
            return;
        }
        hlp = list_entry(history.tail.prev, struct history_line, elem);
        printf("%s\n", hlp->line_str);
        runcmd_builtin( cmd_str: hlp->line_str);
        return;
    }

    // Handle !<number>
    if(buf[0]=='!' && buf[1]>='0' && buf[1]<='9'){
        index = atoi(&buf[1]);
        for(e = history.tail.prev; e != &history.head; e = e->prev){
            hlp = list_entry(e, struct history_line, elem);
            if(hlp->line_num == index){
                found = 1;
                break;
            }
        }
        if(found){
            printf("%s\n", hlp->line_str);
            runcmd_builtin( cmd_str: hlp->line_str);
        }
    }
}

```

Through history_run function I enable it to read the buffer to get the command on what to do in shell history, once we write \$!n in QEMU, it runs the nth command in the history. !! runs the last command. The traversing is done through the linked list.